4.3.1 Prove the orthogonality relations

$$\sum_{n\in G_N}\chi(n) = \begin{cases}\phi(N) & \chi=\mathbb{1}\\ 0 & \chi\neq\mathbb{1}\end{cases}, \quad \sum_{\chi\in\widehat{G_N}}\chi(n) = \begin{cases}\phi(N) & n=1\\ 0 & n\neq 1\end{cases}$$

Pf: (1) $\sum_{n\in G_N}\mathbb{1}(n) = \sum_{n\in G_N} 1 = \phi(N)$.

For $\chi\neq\mathbb{1}$, $\sum_{n\in G_N}\chi(n) \overset{\chi(n_0)\neq 1}{=} \sum_{n\in G_N}\chi(n_0 n) = \chi(n_0)\sum_{n\in G_N}\chi(n)$

Hence $\sum_{n\in G_N}\chi(n) = 0$. (since $|\chi(n_0)|=1$ $\forall n_0$) $\Rightarrow \chi(n_0)\neq 0$.

(2) $\sum_{\chi\in\widehat{G_N}}\chi(1) = \sum_{\chi\in\widehat{G_N}} 1 = \phi(N)$.

For $n\neq 1$, $\sum_{\chi\in\widehat{G_N}}\chi(n) = \sum_{\chi\in\widehat{G_N}}(\mathbb{1}\chi)(n) = \sum_{\chi\in\widehat{G_N}}\mathbb{1}(n)\chi(n) = \phi(N)\sum_{\chi\in\widehat{G_N}}\chi(n)$.

Hence $\sum_{\chi\in\widehat{G_N}}\chi(n) = 0$.

4.3.2. Let χ be primitive Dirichlet character mod N, $m \in \mathbb{Z}$. Prove $\sum_{n=0}^{N-1} \chi(n) \mu_N^{nm} = \bar{\chi}(m) g(\chi)$ as follows.

(a) For $m=0$, confirm for $\chi = \mathbb{1}_1$ and for nontrivial χ.

Pf: If $\chi = \mathbb{1}_1$, then $LHS = \sum_{\substack{0 \le n \le N-1 \\ (N,n)=1}} \mu_N^{n \cdot 0} = \phi(N)$.

$RHS = g(\chi) = \sum_{n=0}^{N-1} \chi(n) \mu_N^n = \sum_{\substack{0 \le n \le N-1 \\ (N,n)=1}} \mu_N^n$.

In general $RHS = 0$. If $N = 1$, then $RHS = \phi(N)$.

For nontrivial χ, denote $S(m) = \sum_{n=0}^{N-1} \chi(n) \mu_N^{nm}$

$\cancel{LHS = 0 \sum_{(N,n)=1} (\mu_N^m)^n = \phi(N)}$.

$LHS = \sum_{(N,n)=1} (\mu_N^m)^n = \sum_{(N,n)=1} \mu_N^{mn}$

$RHS = \bar{\chi}(m) \sum_{(N,n)=1} \mu_N^n = \sum_{(N,n)=1} \mu_N^n$.

(b) i) Now assume $m \neq 0$. Let $g = \gcd(m, N)$, $\begin{cases} N = N'g \\ m = m'gh \end{cases}$

for minimal h such that $\gcd(m', N) = 1$.

Show since $\bar{\chi}(m')\chi(m') = 1$, $\sum_{n=0}^{N-1} \chi(n) \mu_N^{nm} = \bar{\chi}(m') \sum_{n'=0}^{N'-1} \left(\sum_{\substack{n=0 \\ n \equiv n'(N')}}^{N-1} \chi(n) \right) \mu_{N'}^{n'h}$

Pf: $\mu_N^{nm} = e^{\frac{2\pi i nm}{N}} = e^{\frac{2\pi i n m'h}{N'}} = \mu_{N'}^{nm'h}$

Thus $\sum_{n=0}^{N-1} \chi(n) \mu_N^{nm} = \sum_{n=0}^{N-1} \chi(n) \mu_{N'}^{nm'h} = \bar{\chi}(m') \sum_{n=0}^{N-1} \chi(nm') \mu_{N'}^{nm'h}$

(nm' runs through $0, \dots, N-1$) $\bar{\chi}(m') \sum_{n=0}^{N-1} \chi(n) \mu_{N'}^{nh}$ rearrange $= \bar{\chi}(m') \sum_{n'=0}^{N'-1} \left(\sum_{\substack{n=0 \\ n \equiv n'(N')}}^{N-1} \chi(n) \right) \mu_{N'}^{n'h}$.

(4.3.2)

ii) Let $K = K_{N,N'} = \{n \in (\mathbb{Z}/N\mathbb{Z})^{\times} ;\ n \equiv 1 \ (N')\} \triangleleft (\mathbb{Z}/N\mathbb{Z})^{\times}$
$= \ker(\pi_{N,N'})$.

~~Show~~ Use the fact that χ is primitive to show that

$$\sum_{n \in K} \chi(n) = \begin{cases} 1 & g = 1 \\ 0 & g > 1 \end{cases}$$

Pf: Recall $N = N' g$.

$g = 1 \Rightarrow N = N' \Rightarrow K = 1 \Rightarrow \sum_{n \in K} \chi(n) = \chi(1) = 1$.
trivial group

if $g > 1$, since χ primitive, $\chi_d \circ \pi_{N,d} = \chi_N \Rightarrow d = N$.

That is, $\chi(K) = 1$. Thus $\sum_{n \in K} \chi(n) = |K| \cdot 1 = \frac{|G_N|}{|G_{N'}|}$

iii) Since $(\mathbb{Z}/N\mathbb{Z})^{\times} = \bigcup_{n'} n'K$, where n' (N') run through $(\mathbb{Z}/N'\mathbb{Z})^{\times}$
coset representatives

show the inner sum in (4.14): $\sum_{\substack{n \geq 0 \\ n \equiv n' (N')}}^{N-1} \chi(n) = \sum_{n \in n'K} \chi(n)$.

Pf: This is to compare $n'K$ with $\{n : \begin{array}{l} n \equiv n' (N') \\ 0 \leq n \leq N-1 \end{array}\}$.

iv) That is $\begin{cases} \chi(n') & g = 1 \\ 0 & g > 1. \end{cases}$

Pf: $\sum_{n \in n'K} \chi(n) = \sum_{m \in K} \chi(n')\chi(m) = \chi(n') \cdot \begin{cases} 1 & g = 1 \\ 0 & g > 1. \end{cases}$

v) Show (4.11) follows in both cases.

Pf: in (4.14), RHS $\overset{g=1}{=} \bar{\chi}(m') \sum_{n'=0}^{N^*-1} \chi(n') \mu_{N^*}^{n'h} = \bar{\chi}(m') \bar{\chi}(h) \sum_{n'=0}^{N-1} \chi(n')\chi(h) \mu_N^{n'h} = \bar{\chi}(m') g$

RHS $\overset{g>1}{=} \bar{\chi}(m') \cdot 0 = 0$. LHS =

4.3.3 (a) Show $\mathcal{M}_k(N, \mathbb{1}) = \mathcal{M}_k(\Gamma_0(N))$.

Pf: $\mathcal{M}_k(N, \mathbb{1}) = \{ f \in \mathcal{M}_k(\Gamma_1(N)) : f[\gamma]_k = \mathbb{1}(d_\gamma) f \ \forall \gamma \in \Gamma_0(N) \}$

$= \{ f \in \mathcal{M}_k(\Gamma_1(N)) : f[\gamma]_k = f \ \forall \gamma \in \Gamma_0(N) \}$

$= \mathcal{M}_k(\Gamma_0(N))$.

(b) Show $\mathcal{M}_k(N, \chi) = \{0\}$ unless $\chi(-1) = (-1)^k$.

Pf: If $f \in \mathcal{M}_k(N, \chi)$, then $f[(-I)]_k = \chi(-1) f$.

$f[(-I)]_k = (-1)^k f$. Thus $f \neq 0 \iff \chi(-1) = (-1)^k$.

(so $\mathcal{M}_{2k-1}(\Gamma_0(N)) = 0$).

4.3.4. (a) Show $\mathcal{M}_k(\Gamma_1(N)) = \bigoplus_\chi \mathcal{M}_k(N, \chi)$.

Pf: By finite group rep theory. G_N 的decomposition.

Another method: $\forall \chi : G_N \to \mathbb{C}^*$

define $\pi_\chi = \frac{1}{\phi(N)} \sum_{d \in (\mathbb{Z}/N\mathbb{Z})^\times} \chi(d)^{-1} \langle d \rangle : \mathcal{M}_k(\Gamma_1(N)) \to \mathcal{M}_k(\Gamma_1(N))$

Then $\pi_\chi^2 = \frac{1}{\phi(N)^2} \sum_{d, d' \in G_N} \chi(dd')^{-1} \langle dd' \rangle = \pi_\chi$.

$\pi_\chi(\mathcal{M}_k(\Gamma_1(N))) \subset \mathcal{M}_k(N, \chi)$.

$\sum_\chi \pi_\chi = \mathbb{1}$, $\pi_\chi \circ \pi_{\chi'} = 0$ if $\chi \neq \chi'$.

Thus $\mathcal{M}_k(N, \chi)$ span and linearly disjoint.

(6.3.4) (b) Show same for $S_k(N,\chi)$.

Pf: $S_k(\Gamma,W) \subset M_k(\Gamma,W)$ is a G_N-subrep.

If G abelian, $W < V$ as G-rep

then $W = \bigoplus_\chi (W \cap V^\chi)$.

Thus $\bigoplus(S_k(\Gamma,W) \cap M_k(N,\chi)) = \bigoplus S_k(N,\chi) = S_k(\Gamma,W)$.

(c) Show same for $\mathcal{E}_k(N,\chi)$.

Pf: By ~~CRT~~, isomorphism thm

$$\bigoplus \frac{M_k(N,\chi)}{S_k(N,\chi)} \simeq \bigoplus \frac{M_k(N,\chi)}{\bigoplus S_k(N,\chi)}$$

$$\wr\wr \qquad\qquad \wr\wr$$

$$\bigoplus \mathcal{E}_k(N,\chi) \qquad \mathcal{E}_k(\Gamma,W) \simeq M_k(\Gamma,W)/S_k(\Gamma,W)$$